Assignment 5.4

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Task 1

```
# Simple Login System (AI-generated example)

username = "admin"
password = "12345"

user_input = input("Enter username: ")
pass_input = input("Enter password: ")

if user_input == username and pass_input == password:
    print("Login successful")
else:
    print("Invalid credentials")
```

Secure

```
import hashlib
import getpass

# Simulated stored credentials (hashed password)
```

```
stored_username = "admin"
stored_password_hash =
hashlib.sha256("12345".encode()).hexdigest()

user_input = input("Enter username: ").strip()
pass_input = getpass.getpass("Enter password: ").strip()

# Hash the entered password
input_password_hash =
hashlib.sha256(pass_input.encode()).hexdigest()

if user_input == stored_username and input_password_hash
== stored_password_hash:
    print("Login successful")
else:
    print("Invalid credentials")
```

Task 2

```
# Fair Loan Approval System

income = float(input("Enter monthly income: "))
credit_score = int(input("Enter credit score: "))

if income > 30000 and credit_score > 650:
```

```
    print("Loan Approved")
else:
    print("Loan Rejected")
```

Output

```
Enter applicant name: Ravi
Enter gender (M/F): M
Enter monthly income: 32000
Enter credit score: 660

Loan Approved
```

Task 3

```
# Recursive Binary Search Program

def binary_search(arr, left, right, target):
    # Base case: element not found
    if left > right:
        return -1

    # Find middle index
    mid = (left + right) // 2

    # If target is found at mid
    if arr[mid] == target:
```

```
        return mid

    # Recursive case: search left half
    elif arr[mid] > target:
        return binary_search(arr, left, mid - 1, target)

    # Recursive case: search right half
    else:
        return binary_search(arr, mid + 1, right, target)


# Main program
numbers = [2, 5, 8, 12, 16, 23, 38, 56, 72]
target = int(input("Enter number to search: "))

result = binary_search(numbers, 0, len(numbers) - 1, target)

if result != -1:
    print("Element found at index", result)
else:
    print("Element not found")
```

Output

```
Enter number to search: 23
Element found at index 5
```

Task 4

```
# Job Applicant Scoring System (AI-generated example)

name = input("Enter applicant name: ")
gender = input("Enter gender: ")
skills = int(input("Enter skills score (0-10): "))
experience = int(input("Enter experience in years: "))
education = int(input("Enter education score (0-10): "))

score = 0

score += skills * 3
score += experience * 2
score += education * 2

# Potentially biased rule
if gender == "M":
    score += 5

print("Applicant Score:", score)

if score >= 50:
    print("Selected")
else:
    print("Not Selected")
```

Output

Enter applicant name: Ravi
Enter gender: M
Enter skills score (0-10): 8
Enter experience in years: 5
Enter education score (0-10): 7

$(8 \times 3) + (5 \times 2) + (7 \times 2) + 5$
$= 24 + 10 + 14 + 5$
$= 53$
Applicant Score: 53
Selected